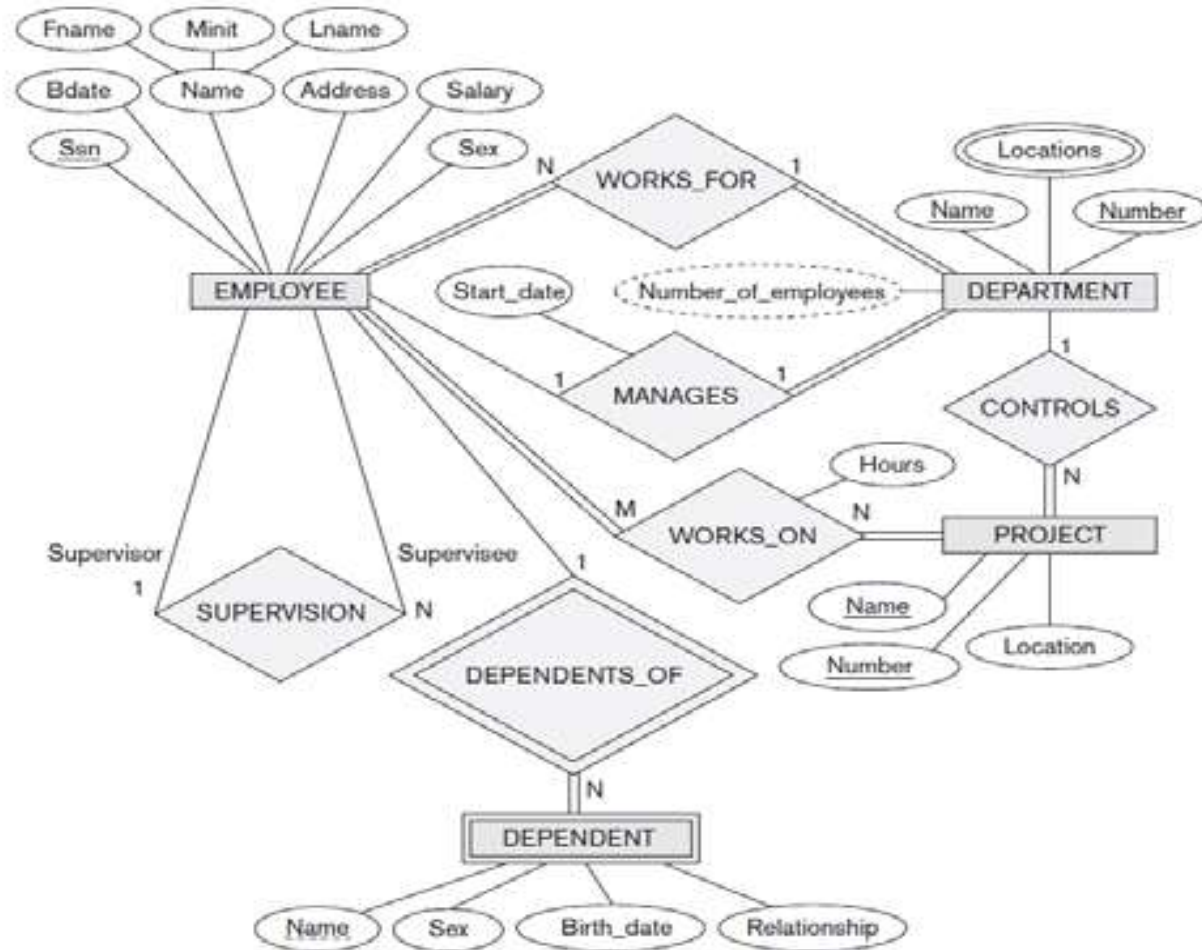


Relational Database Design Using ER-to-Relational Mapping

Figure 9.1

The ER conceptual schema diagram for the COMPANY database.



Sample Mapping of ER Schema to Relational Database Schema

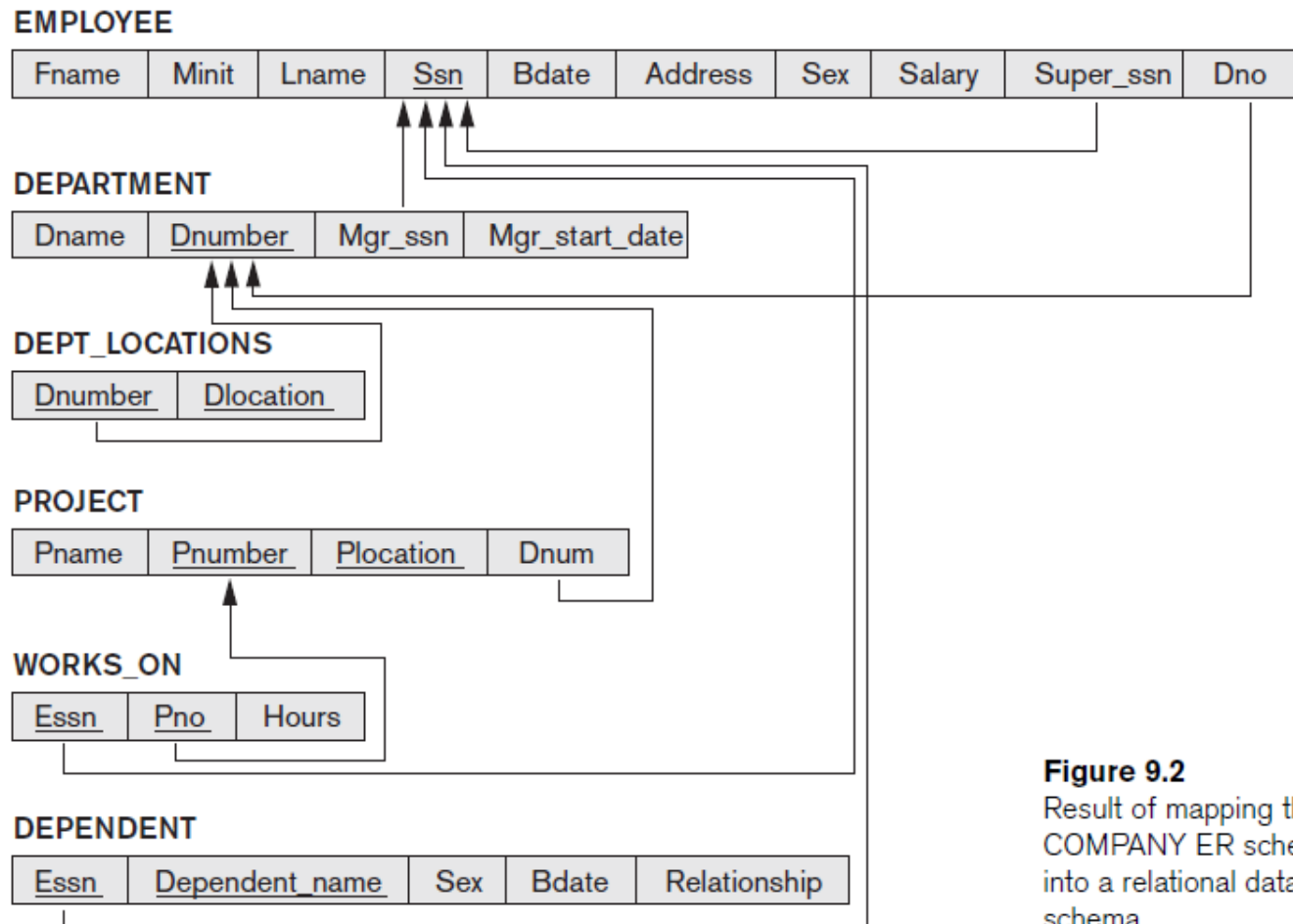


Figure 9.2
Result of mapping the
COMPANY ER schema
into a relational database
schema.

- COMPANY database example
 - Assume that the mapping will create tables with simple single-valued attributes
- Step 1: Mapping of Regular Entity Types
 - For each regular entity type, create a relation R that includes all the simple attributes of E
 - Called **entity relations**
 - Each tuple represents an entity instance

- Step 2: Mapping of Weak Entity Types
 - For each weak entity type, create a relation R and include all simple attributes of the entity type as attributes of R
 - Include primary key attributes of owner as foreign key attributes of R

(a) **EMPLOYEE**

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary
-------	-------	-------	------------	-------	---------	-----	--------

DEPARTMENT

Dname	<u>Dnumber</u>
-------	----------------

PROJECT

Pname	<u>Pnumber</u>	Plocation
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(b) **DEPENDENT**

<u>Essn</u>	<u>Dependent_name</u>	Sex	Bdate	Relationship
-------------	-----------------------	-----	-------	--------------

(c) **WORKS_ON**

<u>Essn</u>	<u>Pno</u>	Hours
-------------	------------	-------

(d) **DEPT_LOCATIONS**

<u>Dnumber</u>	<u>Dlocation</u>
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- Step 3: Mapping of Binary 1:1 Relationship Types
 - For each binary 1:1 relationship type
 - Identify relations that correspond to entity types participating in R
 - Possible approaches:
 - **Foreign key approach**
 - **Merged relationship approach**
 - **Crossreference or relationship relation approach**

- Step 4: Mapping of Binary 1: N Relationship Types
 - For each regular binary 1: N relationship type
 - Identify relation that represents participating entity type at N -side of relationship type
 - Include primary key of other entity type as foreign key in S
 - Include simple attributes of 1: N relationship type as attributes of S
 - Alternative approach
 - Use the **relationship relation** (cross-reference) option as in the third option for binary 1:1 relationships

Step 5: Mapping of Binary $M:N$ Relationship Types

- For each binary $M:N$ relationship type
 - Create a new relation S
 - Include primary key of participating entity types as foreign key attributes in S
 - Include any simple attributes of $M:N$ relationship type

Step 6: Mapping of Multivalued Attributes

- For each multivalued attribute
 - Create a new relation
 - Primary key of R is the combination of A and K
 - If the multivalued attribute is composite, include its simple components

- Step 7: Mapping of N -ary Relationship Types
 - For each n -ary relationship type R
 - Create a new relation S to represent R
 - Include primary keys of participating entity types as foreign keys
 - Include any simple attributes as attributes

Table 9.1 Correspondence between ER and Relational Models

ER MODEL	RELATIONAL MODEL
Entity type	<i>Entity</i> relation
1:1 or 1:N relationship type	Foreign key (or <i>relationship</i> relation)
M:N relationship type	<i>Relationship</i> relation and <i>two</i> foreign keys
<i>n</i> -ary relationship type	<i>Relationship</i> relation and <i>n</i> foreign keys
Simple attribute	Attribute
Composite attribute	Set of simple component attributes
Multivalued attribute	Relation and foreign key
Value set	Domain
Key attribute	Primary (or secondary) key